



QUEZON CITY UNIVERSITY



**PATIENT INFORMATION SYSTEM FOR BARANGAY GULOD
HEALTH CENTER**

A Capstone Project Documentation

Presented to

The College of Computer Science and Information Technology

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In Partial Fulfillment

of the Requirements for the Degree

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

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APPROVAL SHEET

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DEDICATION

This research is lovingly dedicated to our beloved and respected parents who have always been our constant source of inspiration and strength. The one who continually gave us their enthusiasm, moral, spiritual, emotional, and financial support. Without them, this research would not have been made possible.

We also dedicate this research to our mentor, friends, and classmates who shared their thoughts, advice, and encouragement to finish this study.

And most of all, we dedicate this study to the Almighty God who gives us protection, vigor, and sound judgment to accomplish this research excellently.



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EXECUTIVE SUMMARY

Title : Patient Information System for Barangay Gulod Health Center

Proponents :

Mentor :

Degree : Bachelor of Science in Information Technology

Date Completed : June 2022

Keywords: Patient Information System, Web Based, Information System.



The Patient Information system is designed for the health center of Barangay Gulod to provide high quality medical records, particularly in medical histories that allow the users quickly access the records or the information needed. Patient Information System is a Web and Lan based to access services such as patient information recording and tracking the number of cases of diseases like dengue and measles for the Barangay Gulod Health Center, the system aimed to improve the residents information in terms of collecting, management, and securing a data.

The main objective of this study is to help lessen the burden of the patients of Gulod and for the health center that will handle the patient data in order to address to create a system to manage patient data in order to address the current system's problem, which is that each patient's data is manually kept, difficult to access, and takes a long time to locate; as a result, patient queues are growing. Using the system's archive features, Barangay Gulod would be able to access all of their patient's data. The goal of the project is to create an accurate, user-friendly system that will benefit patients in Barangay Gulod.



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CHAPTER I

CAPSTONE PROJECT BACKGROUND

INTRODUCTION

A patient record is a collection of records that detail each episode in which a patient sought and received treatment or a referral from a healthcare facility. The information in the record is kept confidential and usually held by the facility, and it is only disclosed to the patient or with the patient's explicit permission. It includes the initial assessment of the patient's health status, the patient's medical history, and other relevant information. In addition, an issue-oriented medical record contains the main problem list. The patient record is usually a stack of papers, but it can also be computerized.

The patient record and information system are designed for medical practitioners and patients to provide high-quality medical records, particularly medical histories that allow system users to quickly access the information they need, as well as to help retain data accurately and improve data control. The technology gives a quick way for the nurse to input recommendations, medicine, and other related information. The basic or standard procedure for documenting medical history is a lengthy process; conventional recording of the system often consists of papers, medical sheets, order sheets, and a collection of documents kept



in a folder. A current means of documenting or maintaining patient medical histories or information is in high demand in our generation today. As technology advances, new methods of recording or storing medical histories emerge. It also defines a system's entire exterior behavior. As part of establishing a thorough information description, detailed functional description, indication of performance requirements, and the design, the software is assigned function and performance. Also, it offers relevant information about the needs.

The proponents proposed a Patient Information System for Barangay Gulod Health Center system that stores patient's information on a database for backup safety in case of information loss and a convenient storing of data and effortless searching of Patient profiles. The Barangay Gulod Health Center's Patient Information System has a function that alerts patients when their doctor requires a re-consultation.



Project Context and Its Background

The Novaliches, Quezon City's residential district includes Gulod, which means "Uphill" and the current Barangay captain is Rey Aldrin S. Tolentino. Barangay Gulod Health Center is a government-run health center in Metro Manila's Quezon City located at #27 Villaflor Village, Brgy. Gulod, District 5. Paper-based registration and recording of information are unrecoverable once they have been lost. Electronic records, on the other hand, include data backup and storage, so there's still a chance for the organization to retrieve the information if it was compromised by a harmful party. When sensitive data is stolen or leaked, it may result to lack of confidence and trust to the health care system. Having well established security of storing patient's information helps the organization is fundamental in providing good health services to the clients. Health care organizations can use a Patient Information System to collect, store, and manage patient treatment records and other sensitive files to lower the risk of errors and loss of medical records.

The goal of the Patient information system, may be incredibly beneficial to the proponents and healthcare staff by allowing them to make better decisions and improve the health service outcomes. Gulod Health Center has encountered many problems with their existing system like information loss, human error, and disorganized information that is being written down on their logbook. Finding old



patient record can also be time-consuming. There is also a higher risk of inaccurate information, misguided application forms, data inconsistency, along with slow and ineffective process of medical check-up, risk of privacy and security, high costs, unreadable handwriting, shortage of registration forms, and difficulty checking the status of the application.

To address this problem, the proponents developed a Patient Information System for Barangay Gulod Health Center. Improving the patient records could make a significant contribution to the barangay's health care system.

Project Purpose and Description

Respondents can use this system to access services such as patient information recording and tracking the number of cases of diseases like dengue and measles for the Barangay Gulod Health Center. The proposed system aims to improve resident information collecting, management, and security. The proponents will be providing a Patient Information System that allows the storage, recording, updating, viewing, editing, and securing of a patient's records, as well as their management. One of the most crucial aspects of a patient information system is security, because it accesses, analyzes, or stores large amount of sensitive data on a daily basis. To put it another way, the system will benefit both the doctors and the patients. The purpose of this study is to develop



that will handle patient data in order to address the present system's problem, in which each patient's data is manually stored, difficult to access, and takes a long time to locate; As a result, patient queues are expanding. It will help the Barangay Gulod to retrieve every data of their patient using the archive features of the system. The project's purpose is to have accurate, easy navigation, and a userfriendly system that will help the Barangay Gulod patients.

The objective of the Study

Main Objective

The main objective of this project is to develop a Patient Information System for Barangay Gulod Health Center.

Specific Objectives

The specific objectives of the project are the following:

1. To develop a system that can manage patient information.
2. To develop a system that has a live chat via Messenger that can be managed during operating hours.
3. To develop a system that will allow users to generate and print reports.
4. To develop a system that has four(4) access levels namely, the administrator, health professionals, assistance desk, and the patients.



5. To develop a web application that has patient modules that can show basic information about the patient's consultations.
6. To develop a web application that has a family tab.
7. To develop a system that can download back-up online.
8. To develop a system that can create and post announcement.
9. To evaluate the Patient Information System for Barangay Gulod health center base on the following ISO25010 criteria:
 - Functionality
 - Reliability
 - Usability
 - Efficiency
 - Security



Scope and Delimitation of the Project

The project is composed of the following:

Scope

1. The system can add, edit, and archive patient information.
2. The system includes a web-based feature that responds to frequently asked questions regarding the website. The system has graphs about the disease with patient age to monitor the number of cases daily, weekly, monthly, and yearly.
3. The system has an online database for easy control and sharing via the internet.
4. Doctors can set a follow-up check-up appointment for patients and also can view details of previous consultations and the upcoming schedule of patients.
5. The system has a consultation form for the Doctor's Module that allows the doctors to input the complaints and diagnoses of patients.
6. Patients can view their profile, their follow-up check-up schedule, and consultation history that may also include patient complaints and doctor's findings.
7. Patients with account can register their other family members such as babies and older relatives in the family tab.
8. The system has an appointment history.



9. The system includes database backup feature, in which the system downloads database from an online database and saves it into an excel file.
10. System administrators can easily notify visitors by creating an announcement and posting it on the website.

Delimitation

1. The system is intended only for Barangay Gulod Health Center.
2. The system does not have an online scheduling feature.
3. The system does not allow doctors on call and testing laboratories.
4. The system does not allow medical certificates, letters, ledgers, and any other medical financial assistance.



Theoretical Framework and Conceptual Framework of the Study

Theoretical Framework

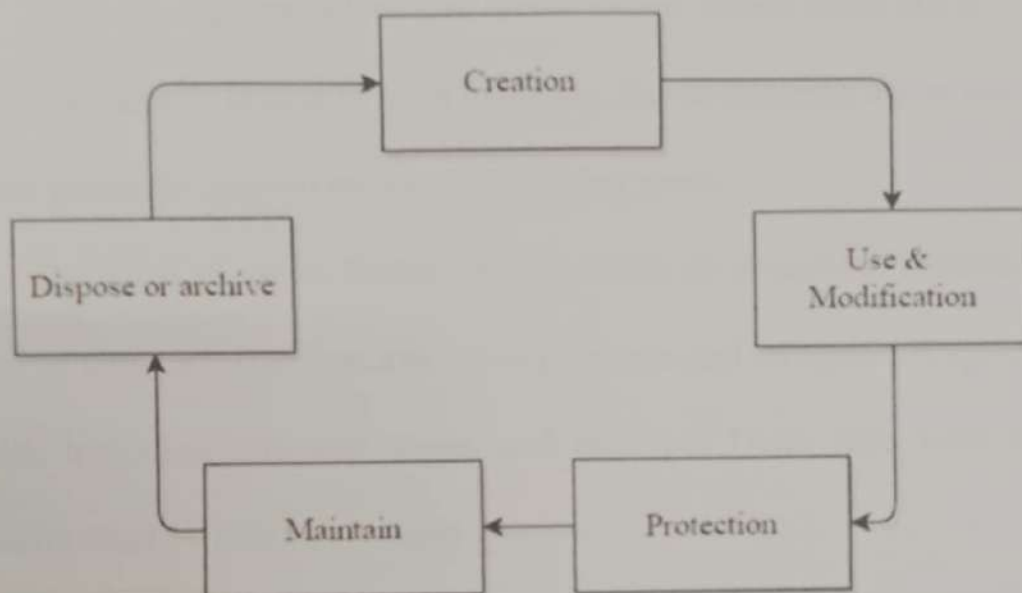


Figure 1.1: Records Management Life Cycle

Resource Data Management, Inc. (2020) There is a lifecycle in records management that correlates to the stages that each record goes through. Everything from the creation of a record until its disposal is covered by its lifetime. Each phase has its own set of policies and processes. It starts with the creation of document and ends with its disposal or preservation. The phases may have different names in different programs, software, and educational materials, but they are all fixed and work simultaneously and continually.

The first phase in a record lifecycle is Creation. It involves the receipt of a form. Classification of it as a record in an organizational record management system is the first phase in a record lifespan. Make sure you are creating records



appropriately, including the relevant information and following the correct structure.

The second phase is Use and Modification. When users utilize and change an active record, it is said to be in this state. Dynamic forms are in use right now and are generally close to the people utilizing them.

The third phase is Protection. A record or document continues to be a record as long as it is in use. You must preserve and safeguard it against various threats, including unlawful access and damage. Those who need documents regularly must be able to get them quickly.

The fourth phase is Maintaining. The amount of time you must keep a record depends on several things, including company policy and government norms and regulations. You will need to keep some records indefinitely. Thus, you need to start an archival procedure at some time.

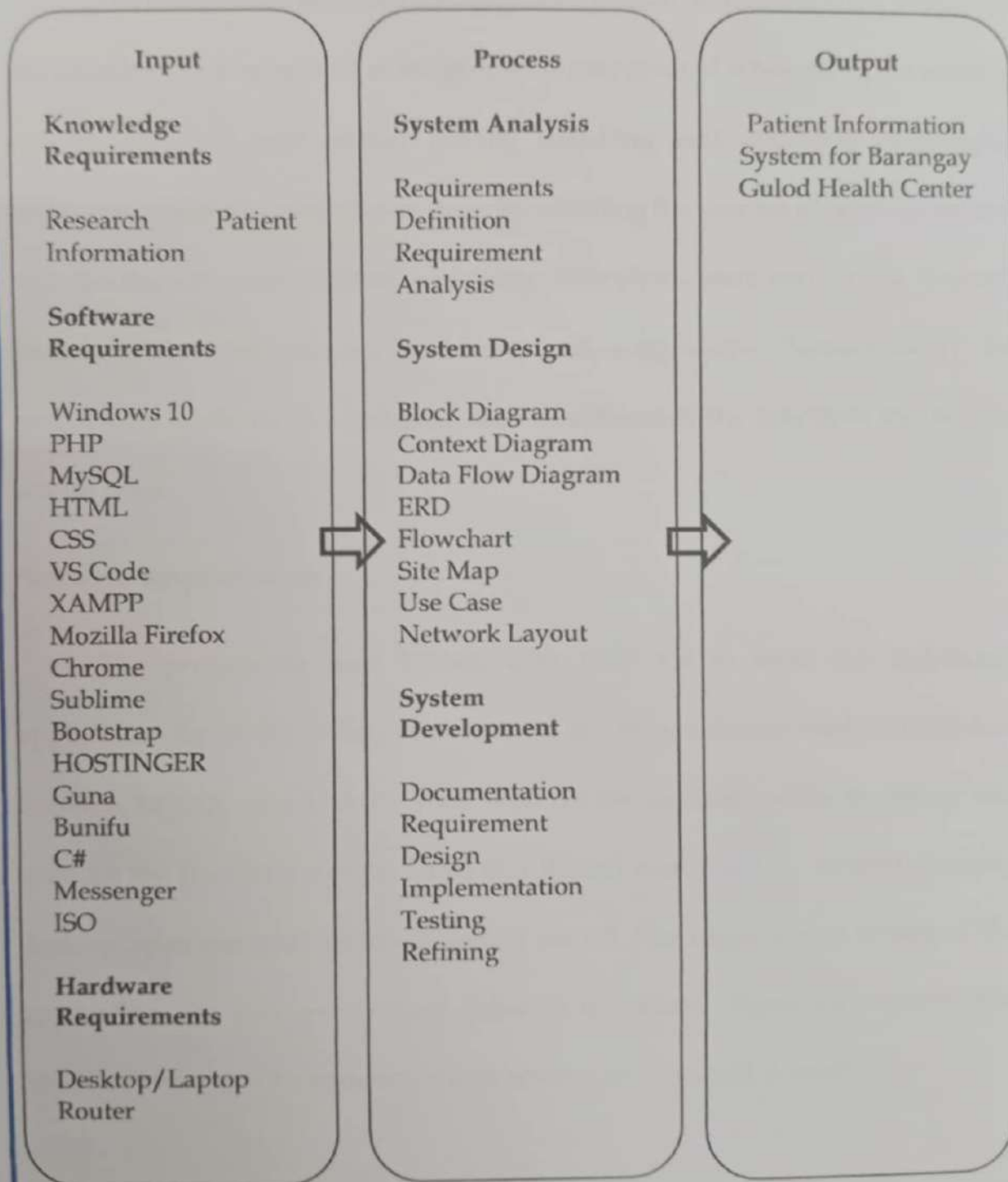
The last phase is Dispose and Archive. The records management team must determine whether to discard or maintain a record when it reaches the end of its lifecycle.

Because most of the records that will enter the system are confidential, the study could benefit from this notion by using it as a guide on handling and maintaining records appropriately.



Conceptual Framework of the Study

Figure 1.2: IPO Model for the Proposed Patient Information System for Barangay Gulod Health Center





Input

Knowledge Requirements

The proponents gathered data and information to determine the process of the proposed system as well as the problems encountered while using the existing system. Research was carried out by searching and obtaining meaningful resources, while observation was done by watching the process of how the patient and the doctors communicate, and lastly, interviews were carried out through inquiry about the process, problems, and suggestions. Subsequently, the proponents analyzed the gathered data to determine the functions the system would have.

Software Requirements

The proponents used HTML, CSS, PHP, C# to build the web-based application. As for the coding environment, the programmers used VS Code and Sublime. MySQL and XAMPP were used for the database while Bootstrap was used for the front-end design. Guna and Bunufi were used to craft stunning desktop apps using UI which is perfect for C#. For running and testing of the application, the proponents used different browsers. These are Chrome and Mozilla Firefox and Hostinger for web hosting services and domain.



Hardware Requirements

To develop and run testing on the system, the proponents used both laptop and desktop with internet connection. The said hardware devices are essentials to run the software needed for the development such as IDEs, browsers, and databases.

Process

System Analysis

In developing the system, the proponents researched and analyzed the key processes and functions of the system, including the problems encountered in using the system, and the solution to these problems.

System Design

In developing the system, the proponents created diagrams that would serve as a guide for the programmers. The system designers were responsible in the system layout to better visualize the flow and functions.

System Development

In the development of the system, programmers are assigned to program the system. Designers create the design and layout for the designer programmers to integrate it into the user interface of the system. The proponents create many prototypes, tweaking and changing based on the feedback given to the



developers. The system will be tested until it meets the requirements then the developers will change little things that can refine the whole system.

In developing the proposed system, the designers were given the task to create a wire-frame and layout for the system, the analyst was tasked to create needed diagrams, and the programmers were assigned to create a prototype and code the proposed system.

The proponents also tested the system to check for errors and asked for feedback until the expectations for the proposed system are met.



Definition of Terms

Describes or explains technical terminology. Technical definitions are intended to introduce language that allows for concise and unambiguous communication in a certain sector.

Technical

Account - an arrangement by which a user is given personalized access to a computer, website, or application, typically by entering a username and password.

Administrator - a person who manages a computer network or system network administrators.

Application - is a computer software package that performs a specific function directly for an end user or, in some cases, for another application.

Barangay - serves as the primary planning and implementing unit of government policies, plans, programs, projects, and activities in the community, and as a forum wherein the collective views of the people may be expressed, crystallized and considered, and where disputes may be amicably settled.

Browser - a computer program with a graphical user interface for displaying and navigating between web pages.



Check-up - is a medical examination by your doctor or dentist to make sure that there is nothing wrong with your health.

Consultation - a meeting with an expert or professional, such as a medical doctor, in order to seek advice.

Data - is defined as facts or figures, or information that's stored in or used by a computer.

Database - is information that is set up for easy access, management and updating.

Develop - the process of defining, designing, testing, and implementing a new software application or program.

Doctor - a person licensed to practice medicine, as a physician, surgeon, dentist, or veterinarian.

File - A file is an object on a computer that stores data, information, settings, or commands used with a computer program.

Health - Health is a state of complete physical, mental, and social well-being and not merely the absence of disease or infirmity.

Live Chat - Live chat is typically implemented through code that is inserted on selected pages of a website and organizations can decide which pages to make live chat available.



Schedule - is a method that is used to distribute valuable computing resources, usually processor time, bandwidth and memory, to the various processes, threads, data flows and applications that need them.

Operational

Account - a record or statement of financial expenditure and receipts relating to a particular period or purpose.

Application - the action of putting something into operation.

Barangay - a small territorial and administrative district forming the most local level of government.

Browser - is an application program that provides a way to look at and interact with all the information on the World Wide Web.

Check-up - a thorough examination, especially a medical or dental one.

Consultation - a meeting to discuss something or to get advice

Data - facts and statistics collected together for reference or analysis.

Database - a structured set of data held in a computer, especially one that is accessible in various ways.

Develop - grow or cause to grow and become more mature, advanced, or elaborate.

Doctor - a person with a medical degree whose job is to treat people who are sick or hurt:



File - a folder or box for holding loose papers that are typically arranged in a particular order for easy reference.

Live Chat - is technology that provides companies with a way to interact with users when they visit an organization's media properties.



CHAPTER II

REVIEW OF RELATED LITERATURE AND SYSTEMS

This chapter presents the review of local and foreign literature and studies related to the research study being conducted

FOREIGN LITERATURE

Web-based Clinic Management System (CMS)

According to Muhammad, J. and Garba, S., (2019), the complexity of sharing patient records such as prescription detail, referral data, diagnostic status, and appointment schedules between multiple clinic units might be an issue without a fully integrated system. Furthermore, access to medical records necessitates confidentiality. Several studies on healthcare services have been done to address concerns like inconsistency in medical records, a lack of quick record storage and retrieval, and a paper-based approach. As a result, this project aims to construct a web-based application to automate and improve the clinical services of Sule Lamido University Clinic. The program has made it simple for doctors and patients to access electronic records, reducing the expense, difficulty, and ineffectiveness of manually dealing with patients' data and increasing the advantages and profitability of clinic services. It may be utilized to make decisions since it gives comprehensive, trustworthy, accessible, and clear data about the



clinic's progress and limitations. Initially, the present system's faults were discovered; nonetheless, the needs were stated and examined. The system's model was created and developed utilizing the SDLC methodology, and the system was then tested and verified. The findings indicated that the system meets all of the requirements. As a result, the system's intended objectives were accomplished.

Patient Management Software System Features and Benefits

According to Samikshan Sarkar in his blog entitled Patient Management Software System Features and Benefits (2018), within healthcare organizations such as medical clinics or hospitals, patient management systems refer to several efficient automated systems that track patient information, diagnoses, prescriptions, interactions, and encounters. Patient management software raises the bar for patient care excellence. It provides greater economy, improved service quality, greater accuracy, and greater efficiency to healthcare facilities of all sizes. Programs are available to meet a wide range of medical practice requirements.

Online Patient Information System Free Download in Bootstrap and PHP

Based on Online Patient Information System Free Download in Bootstrap and PHP in inettutor website (January, 2022). Developing a website makes it easier to keep track of patient information records. It renders paper-based patient information records and physical storage in hospitals obsolete. The system may



help improve the accuracy of patient information records. The system improves patient information security and privacy. Furthermore, it makes the retrieval of medical records easier and faster for hospitals. Overall, the technology makes managing hospital medical records easy yet more efficient.

Hospital Management System (HMS)

According to the blog of Sanjana entitled, a detailed view of Hospital Management System (HMS), (2018). HMS allows you to manage all of your paperwork in one place, saving your staff time organizing and analyzing patient paperwork. HMS maintains the patient's medical records and contact information, keeps track of appointment dates, saves insurance information for future reference, and tracks bill payments, among other things. Appointment Management, Prescription Management, and Master Information Systems are features of this system that are related to the proposed system of the proponents.

The Importance of Medical Records Management

According to The Importance of Medical Records Management by Joe Weller (2021), a medical record can benefit both the doctor and the patient. The record can outline the patient's history and treatment plan in an easily accessible format. It can provide assurance to the physician about the efficacy of the treatment plan. If someone needs to check a test, or medication, or make a quick



medical decision, a good medical records management system can mean the difference between life and death. Medical records management can include anything and everything about a practice and a patient, such as a patient's history, clinical findings, diagnostic test results, pre and post-operative care, patient progress, and medications.

The Components of a Medical Record

Based on the blog of Gsimon (2021) titled The Components of a Medical Record, it contains information about a patient's identification, health status, personal and family medical history. The level of detail, amount of information, and type of information greatly differ between patients. The amount of care a patient requires will largely determine their medical documentation. Every patient has medical documentation; therefore, it is important to be well-informed on what is in your medical records, as well as your rights to these vital documents.

LOCAL LITERATURE

Community Health Information Tracking System (CHITS)

According to Development Academy of the Philippines (October, 2020), The Community Health Information Tracking System (CHITS), as the first



homegrown EMR system in the Philippines, employs an open-source, user-friendly, modular, and extensible system that enables the automation of core processes in health centers and thereby contributes to the effective and efficient delivery of services. CHITS is a digital health application that assists health centers and rural health units (RHUs) in improving the delivery of health services by facilitating efficient data entry and storage, facilitating secure access to records by healthcare workers, expediting patient record retrieval, and reducing patient waiting time in clinics and RHUs. It assists health program administrators and local chief executives with resource allocation methods and organizational growth by assisting with patient load projections and resource use. It also provides a more efficient system for arranging appointments and follow-ups, as well as a more responsive way of monitoring the community's health through the data it collects on a daily basis. Furthermore, because CHITS enables timely data entry at the local level, it has become easier for Municipal and City Health Offices (M/CHOs) to generate the necessary health reports for integration into the Field Health Service Information System (FHSIS), the Philippine Department of Health's official reporting and recording system. Because RHUs have access to up-to-date, on-the-ground data, they may use inter-health center integration within their authority to better follow patients. This integration of community health data helps them to forecast logistical demands, allowing them to distribute material and human resources more efficiently and effectively at the local level. Furthermore, CHITS



improves its capacity to monitor community health and detect prospective epidemics on a frequent basis.

Going digital to enhance UHC implementation: health care expert

According to Montemayor Ma. T., (2019) ,a health care expert, healthcare facilities going digital is critical to the success of the Universal Health Treatment (UHC) Law implementation since it allows more patients to receive better care. On the side line of a health forum in Mandaluyong City, St. Luke's Medical Center President Arturo de la Pena told the Philippine News Agency (PNA) that the use of electronic medical records (EMR) would enable hospitals, both private and public, to provide faster and higher-quality health care services, especially now that every Filipino is entitled to them under the UHC law. The electronic medical record (EMR), a relatively new technology used in some Philippine hospitals, connects current hospital systems and offers immediate access to patient information.

EMR in the Philippines amidst the COVID-19 Pandemic

In the Philippines, electronic medical records, or EMR, are being introduced, transforming the way doctors and nurses handle patient care and information. EMR systems are safe and convenient for clinicians to use at any time and from any location in the country. They are also important for getting correct and timely patient information before starting any therapy or surgery. Electronic



medical record (EMR) systems, according to Salon, P. (2021), are more than just a digital record of duplicates of your patients' paper charts because they help you save time and money, eliminate chart loss or misfiling, and provide readable chart information. They provide crucial information that aids clinical decision-making.

The Need for a Patient Records Management System in the Philippines

According to Seymour (2019), as the number of patients increases, the ratio of doctors to patients decreases making it difficult for doctors to remember each patient they encounter. All clinicians must use a Patient Records Management System, a legal EMR system that values hospital personnel and patients' notes and workflow. They will be able to leave the clinic early, complete tasks on time, and avoid errors as a result of writing the same prescriptions over and over again. It has also assisted them and changed their lives for the better. It's made such a difference for their patients.

MedProjects Launches EMR for Hospitals in the Philippines

Health Engine, an integrated suite of hospital IT products that comprises an Electronic Medical Records (EMR) platform, a Social Intranet, and a cloud-based Website and Portal, has been released by a healthcare IT firm (Muin, 2020). All of this is aimed at enhancing collaboration and communication, as well as clinical processes and workflows and patient engagement. Health Engine provides several benefits to Philippine hospitals, including improved cost control,



integration with existing hospital IT systems and devices, and the opportunity to tailor the solution to the hospital's specific needs. It allows medical records to be released promptly.

The system includes clinical and nursing workstations, patient profiles and lists, hospital-wide results viewing, patient chart viewing and management, records tracking, supplemental services, and ambulatory care modules.

Philippines to integrate open-source electronic medical record platform for enhanced health programs

According to Ateneo de Manila University, the Philippine Department of Health Region VII is merging with Smart Communications' SHINE OS+ electronic medical record system and the Ateneo Java Wireless Competency Center (AJWCC) with its existing reporting tool to improve the region's health programs. It was introduced in 2011 and allows health units to communicate aggregated medical data directly to the region's Health Console, allowing for easy retrieval of information. The SHINE OS functions as an electronic medical record (EMR), allowing data to be transferred between systems to improve interoperability. It is now possible to avoid the time-consuming and strenuous task of manually sorting through documents.



FOREIGN STUDIES

Chatbot Acceptance in Healthcare: Explaining User Adoption of Conversational Agents for Disease Diagnosis

The researchers used a research model based on UTAUT2 for explaining the acceptance of CA for disease diagnosis. Unified Theory of Acceptance and Use of Technology is used for consumer contexts, such as Health or the specific consumer app Ada, which is the focus in this research. This paper focuses on the diagnosis of diseases and initiating the subsequent treatment. From the individual's point of view, this also includes self-diagnosis on the internet (Bergman et al. 2011). Healthcare is beset by a confluence of rising demand for services and dwindling availability of healthcare personnel. Mobile Health is a concept offered to help healthcare services overcome geographical, temporal, and organizational obstacles. Conversational agents (CA) like Software applications that engage with people using natural language are being created, and some are even capable of diagnosing an illness based on the user's input via a chat interface. (Laumer, et. al., 2019).



Analysis and Design of Web-Based Health Service Information Systems (E-Health), in the Industrial Revolution Era 4.0

This study intends to give quick service information to the public and can help health care institutions play a bigger role in the community by providing information about health problems that arise in the neighborhood. Especially in the fourth industrial revolution, often known as the digital era, when involvement and use of computer technology are inextricably linked in many aspects of life. One of the health care centers at the sub-district level in Community Health Centres (Puskesmas). According to the study, the health service information system is a vital component that cannot be separated from the country's health system. The vast number of patients makes it difficult to obtain information, total income, and drug supply data utilized by the puskesmas (government-mandated community health clinics) The system development has able to assist the Public Health Service Center in managing current health services, and obtaining essential health information. Officers can swiftly acquire patient data and prescriptions which saves time and resources because they no longer have to dig through ledgers and stacks of maps to find a specific patient health information (2021, Wulan et al.).



On-patient medical information encoding: image guided fistula cannulation assistant for hemodialysis

The research is attested through the creation of a prototype medical device system that tackles clinically reported difficulties with dialysis treatment for patients suffering from kidney failure. The medical system, which is implemented as a temporary tattoo, is designed to be widely deployable in a variety of contexts for a variety of populations, including existing and emerging medical communities, in-clinic or at-home, and across a range of human skin tones. It is critical to address hemodialysis issues since, if one's kidneys fail, the best alternative is to have a transplant, but for many people (millions worldwide), hemodialysis is the only short-term and frequently long-term option. Through this treatment, a patient regularly has large needles inserted into his or her arm through which the blood is pumped into an artificial kidney machine. This must be done several times each week for four to eight hours at a time, often in a clinic though ideally (albeit uncommonly) at the patient's home. The development process for image-guided systems for hemodialysis needle insertion is described, including problem identification, concept generation, testing, and prototyping. This serves to curb fear, thus, patients and caregivers experience improved. Key performance indicators including: less pain as fewer needle are stucked, quicker learning, shortened treatment training time, simpler treatment, reduced



dependence on one or more caregivers when cannulating, and improved access to care.

Patient data representation for outcome prediction of congestive heart failure patients

In the statement of Subramanyan et.al, (2019, Artificial Intelligence (AI) has its roots in every field in the present scenario. The tremendous increase in healthcare data availability and considerable growth in big data analytic methods have paved the way for the success of AI in healthcare and research that significantly improved the quality of service. Electronic Health Records (EHR), which contain temporally ordered patient information, are used to record healthcare data. Heterogeneity, missing values, biases, noise, temporality, and other issues plague EHR data.

Identification & Visualization of Patient Information Elements to Support Chronic Illness Care: A Scoping Review And Pilot Study

Based on the article of Kinch et.al., (2017) the literature about the use of Clinical Information Systems (CIS's) to support the information needs of individual health care providers (HCP), especially the nurse case manager and the interprofessional team that provides chronic illness care in the community. Patient information elements visualization formats, techniques, and organization



and visualization of patient information elements. The most often mentioned information elements were diagnostics and observations. Health care providers (HCP) must attempt to consult various folders and files in order to obtain the complete relevant patient information. This is time-consuming, ineffective, and error-prone. Duplicate data acquisition occurs as a result of ineffectively structuring and displaying pertinent patient information none the clinical record, which is inconvenient for both the patient and the doctor.

Patients, Preferences, and Portals: Barriers Identified to Accessing Personal Health Information through a Secure Online Website

According to Fox-McCloy et.al, (2017) Patient engagement is one of the 6 quality directives issued by the Institute of Medicine for patient-centered care. Health care companies are required under federal meaningful use laws to provide patients with a secure internet platform, known as a patient portal, through which they can access their health information. Even though the patient portal allows people to be more active in their care, it is not frequently used. Barriers to utilization, on the other hand, are best understood from the patient's perspective. Any barriers to patients accessing the portal are also barriers to patient engagement. The purpose of this project was to understand from the patient perspective why 99% were not using the portal in a large health system. The purpose was to learn about patient preferences and expectations for the portal, as



well as perceived roadblocks. This quality improvement project uses the Diffusion of Innovation Theory to better understand the patient perspective and implement targeted portal and program modifications. Patients were interviewed using a focus group method to learn about the portal, their desire to use it, and their overall preferences for accessing health information. With a total of 15 participants, four focus groups were held. Each session was recorded, transcribed using the NVivo application, and then analyzed using content analysis. The most significant impediment to patient portal adoption is a lack of understanding regarding its purpose, utility, and accessibility. Education and promotion materials were proposed as feasible solutions by the participants. Patients will also require information on how to access and use the portal, which will be provided by the nursing staff. Patients will have improved access to their personal health information thanks to the updated site, which will result in positive societal change.

Influence of Mechanism Of Patient-accessible Hospital Information System Implementation On Doctor-Patient Relationships: A Service Fairness Perspective - Science Direct

In the statement of ChangyongLiang et.al, (2017) the relationship between a doctor and a patient plays an important role in patient satisfaction with



healthcare services. In China's healthcare reform, it has become a major and contentious subject. The impact of implementing a patient-accessible hospital information system (HIS) on doctor-patient relationships and patient satisfaction is investigated in this study using service fairness as a theoretical lens. Patient-accessible HIS implementation is linked to perceptions of service fairness, doctor-patient relationships, and patient satisfaction, according to a research model created by us. Data were collected from patients in one of the biggest hospitals in East China that has implemented patient-accessible HIS. Based on the findings, patients' perceptions of service fairness has improved, doctor-patient relationships has strengthened, and patient satisfaction has increased.

LOCAL STUDIES

Patient Management Information System for the University of the Immaculate Conception College Department Clinic

Based on the study of Juvie Pauline L. Relacion, (2017) the patient management information system for the University of the Immaculate Conception College Department Clinic. The goal of the study was to develop a simple system for storing and maintaining patient records and inventory. It creates an automated system to track patients information and medicine inventory, as well as a database for patient information records and medicine inventory reports. It is required because paper-based records demand a large amount of storage space, which can



be easily misplaced or misfiled. Paper work does not only require a lot of time to manage, it also lacks confidentiality and prone to error. The iterative and incremental model was adopted in this investigation. The main idea is to build an application in small chunks over time using repeated cycles.

The findings aid clinic staff in providing healthcare services to patients with more efficiency. The team was also able to reduce any unpleasant patient complaints with the use of this system. Furthermore, the staff was able to meet the administrator's criteria due to the system's ability to provide printed output for patient records and medicine inventory reports.

A Web-Based "InstaSked" Appointment Scheduling System at Perpetual Help Medical Center Outpatient Department

Every hospital concerns about patient pleasure and comfort, according to Mendoza, S., et al (2018). Due to long waiting time, patients have become dissatisfied with the traditional appointment system. This research resulted in the creation of "InstaSked," a novel web-based appointment scheduling system that has the potential to reduce waiting time of patients. Patients (appointment scheduling), medical secretaries (patient lists), physicians, and management will benefit from it (monitoring patients).

Health Emergency and Public Involvement in the Philippines: Syndromic Surveillance Efforts and System Integration



Estuar et al. (2020) found that introducing electronic medical records would be tremendously advantageous to the country. The Philippine Republic, through the Department of Health (DOH) has mandated that all rural health units digitize health records, select and install an electronic medical record (EMR) system. The digitalization of primary care health data is the first step toward universal health care in the country, as well as a more efficient way of accessing health information during times of crisis.

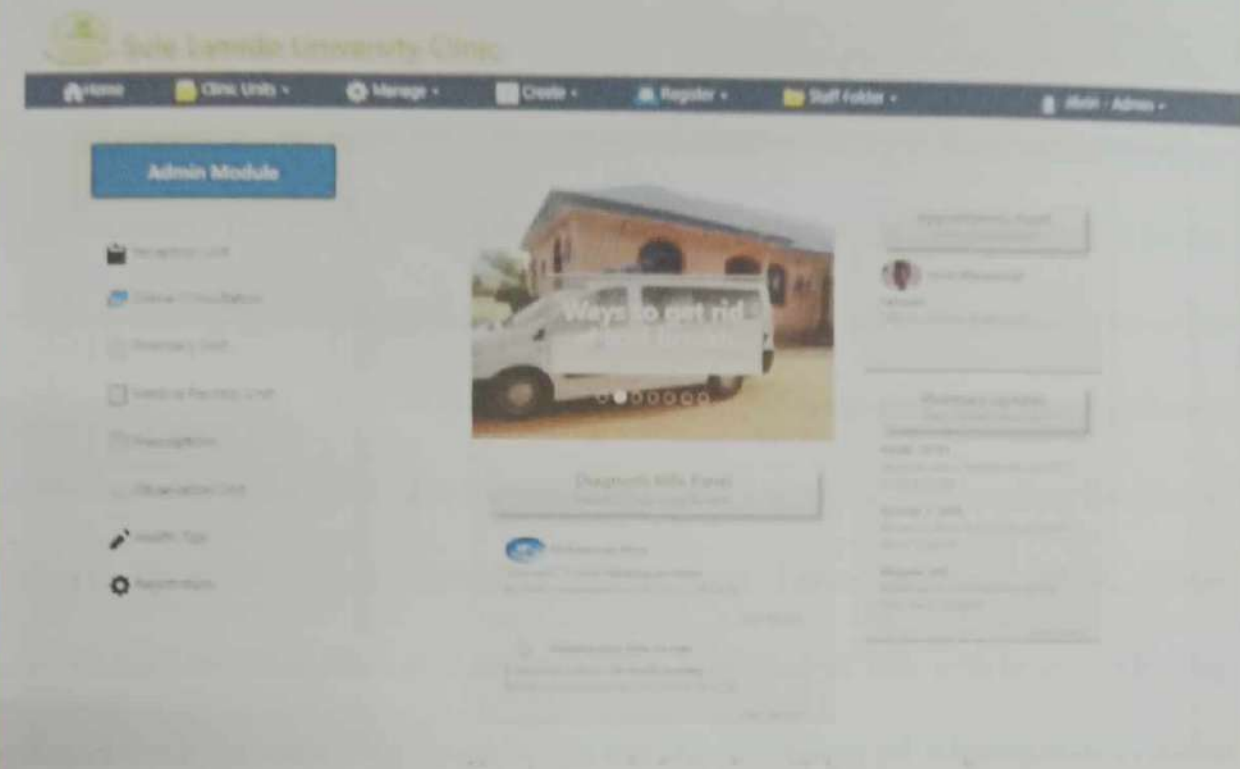
Identifying Healthcare Information Systems Enablers in a Developing Economy

The World Health Organization (WHO) encourages emerging countries to embrace HIS, citing evidence of improved service delivery and efficiency, according to Ebarido and Tuazon (2019). Industrialized economies recognize the need for technological integration in healthcare with national regulations such as the Health Insurance Portability and Accountability Act (HIPAA) and the Personal Health Information Protection Act (PHIPA). Institutions must thrive in the burgeoning health industry if the Philippine healthcare system is to have a future. An institution can improve its ability to provide high-quality care and gain a competitive edge by employing HIS impact.



RELATED SYSTEM

Web-based Clinic Management System (CMS)



SYNTHESIS

With the above-mentioned studies, most of the gathered study and literature is about the Barangay Gulod Patient Information System that is relevant and related to the project. One of the related studies about the Barangay Patient System is in the study of Lahar Mishra (2019) in their study titled Management Information system. He stated that Organizations find it difficult to manage the information as a whole before computer technology bloomed. Developments in computer technology made it possible for the managers to easily gather, integrate,



store and manage the information in the form they require depending upon their needs and timing. The information are used simultaneously by many people.

Record information is something that represents proof of existence and that cannot be used to recreate a proven state of existence, regardless of medium or characteristics. The study proves that the Barangay Information system will be the proof of existence information; it will serve as a platform to determine the other information needed. Most of the related studies and literature that were gathered for the website were indeed a requirement for a Barangay Information System as claimed by João Vidal Carvalho (2016) and Lahar Mishra (2019). These instruments are used also in hospitals. The objective of this article is to identify and compare the maturity models for the management of information systems and technologies (IST) in healthcare.

Technical Background

SOFTWARE

The proponents defined and discussed in this part the technical specifications of the hardware and software requirements that were used in developing and implementing the Patient Information System for Barangay Gulod Health Center. Requirements must be established in a systematic way to ensure the system accuracy and completeness. Requirement analysis is the first



initiative within the system design process, where a user requirement should be clarified and documented to get the corresponding specifications.

Table 2.1 Software Development Requirements

Software	Specification	Descriptions
Operating System	Windows 10	An operating system that is generally common nowadays and is flexible with any program.
HTML, CSS, PHP		HTML is the standard markup language for documents designed to be displayed in a web browser. Cascading Style Sheets is a style sheet language used for describing the presentation of a document written in a markup language such as HTML. PHP is a general-purpose scripting language geared towards web development.
MYSQL, XAMPP		MySQL is an open-source relational database management system.



		XAMPP is a free and open-source cross-platform web server solution stack package developed by Apache Friends, consisting mainly of the Apache HTTP Server, MariaDB database, and interpreters for scripts written in the PHP and Perl programming languages.
Visual Studio Code		Visual Studio Code is a source-code editor made by Microsoft for Windows, Linux, and macOS. Features include support for debugging, syntax highlighting, intelligent code completion, snippets, code refactoring, and embedded Git. Users can change the theme, keyboard shortcuts, preferences, and install extensions that add additional functionality.
Mozilla Firefox		Mozilla Firefox or simply Firefox is a free and open-source web browser developed by the Mozilla Foundation and its subsidiary, the Mozilla Corporation. Firefox uses the Gecko



		rendering engine to display web pages, which implements current and anticipated web standards.
Google Chrome		Google Chrome is a cross-platform web browser developed by Google. It was first released in 2008 for Microsoft Windows, built with free software components from Apple Web-Kit and Mozilla Firefox. It was later ported to Linux, macOS, iOS, and Android, where it is the default browser
Sublime		Sublime Text is a commercial source code editor. It natively supports many programming languages and markup languages. Users can expand its functionality with plugins, typically community-built and maintained under free-software licenses. To facilitate plugins, Sublime Text features a Python API.
Bootstrap		Bootstrap is a free and open-source CSS framework directed at responsive, mobile-first front-end web development. It contains CSS- and JavaScript-based design templates for



		typography, forms, buttons, navigation, and other interface components.
C#		C# is an object-oriented programming language from Microsoft that aims to combine the computing power of C++ with the programming ease of Visual Basic. C# is based on C++ and contains features similar to those of Java.
Messenger		Messenger is a FREE mobile messaging app used for instant messaging, sharing photos, videos, audio recordings, and group chats. The app, which is free to download, can be used to communicate with your friends on Facebook and with your phone contacts.
Hostinger		Hostinger, like many other web hosting services, offers shared web hosting. With this most basic hosting tier, your website lives on a server with other websites. In other words, the sites literally share a single server's cost and resources. As a result, shared web hosting is often low-cost web hosting.



Bunifu		Bunifu Framework helps software developers and companies globally create modern, stunning software interfaces and user experience. The product supports WinForms for C# and VB.NET developers. The features are carefully crafted with the developer in mind to provide speed and simplicity. All these at an affordable cost.
Guna		Guna UI is the suite for creating groundbreaking desktop app UI. It is for developers targeting the .NET Windows Forms platform. Guna UI guarantees faster development and improved productivity.

Table 2.1 shows the software requirements that have been used by the proponents. This table will include the specification and description of the software in developing the system.

HARDWARE

As the definition said, Computer Hardware is a technical specification of the computer's components and its capabilities. The processor speed, model,



manufacturer, memory, and even the network adapters are generally part of the hardware specifications.

Software	Specification	Descriptions
Memory	4GB RAM and above	Memory is a device or system that is used to store information for immediate use in a computer or related computer hardware and digital electronic devices.

Table 2.2 Hardware Requirements Specification

Table 2.2. It is the Hardware Specifications that were used by implementing the proposed system. This table includes the specification and description of the hardware in implementing the system.

TECHNOLOGIES

The technologies used in developing this system are also necessary to include as the proponent's guide on developing this project.

**Table 2.3 Technologies Requirements Specification**

Other Technology Used	Specification	Descriptions
PC/Computer/ Laptop	Intel Core i9	A computer is a programmable device that stores retry and processes data.
Router/ Modem	WAN: One 10/100 Mbps Ethernet RJ-45 port with auto MDI/MDIX support. LAN: Four 10/100 Mbps Ethernet RJ-45 ports with auto MDI/MDIX support.	A device that connects two or more packet-switched networks or subnetworks. It serves two primary functions: managing traffic between these networks by forwarding data packets to their intended IP addresses and allowing multiple devices to use the same Internet connection.



Antenna: 4

External

Antennas.

LED

indicators:

Power/SYS.

WAN.

WLAN_2. 4G.

WLAN_5G.

Buttons: Reset.

WPS.

Power: 12V

DC, 1A.

Power

consumption:

5W.



Flash Drive	64GB (SDCZ80- 064G-A75)	portable data storage device that plugs into the USB port of a computer
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Table 2.3. It is the Technologies Specification that was used by implementing the proposed system. This table includes the specification and description of the technologies in implementing the system.



CHAPTER III

DESIGN AND METHODOLOGY

This chapter discusses the Design and Methodology of the hardware and software requirements that were used in developing and implementing the project. Requirements for System indicates the analysis, documentations, diagrams, and data plan that are required to implement and improve the system.

Methodology

The method of research used by the researchers in this study is descriptive method, where the characteristics of the population, analysis, interpretation of collected data and analyzing the habits of the respondents were described. The purpose of employing the descriptive method was to describe the nature of a condition, as it took place during the time of the study, and to explore the cause or causes of a particular condition.

Descriptive studies are observational studies that describe the patterns of disease occurrence to variables such as person, place, and time. They are often the first step or initial inquiry into a new topic, event, disease, or condition. The design used in the study was survey method research that concluded that the sample population answered the questions administered by the proponents through questionnaires. The given questionnaire contained questions that seek to



answer whether the proposed system is more efficient than their current system. To ensure the reliability and efficiency of the conducted interview, it was made sure the questions were constructed properly for the participants to understand them clearly and easily.

Quantitative research is the process of collecting and analyzing numerical data. It can be used to find patterns and averages, make predictions, test causal relationships, and generalize results to wider populations. Quantitative research is a research strategy that focuses on quantifying the collection and analysis of data. It is formed from a deductive approach where the emphasis is placed on the testing of theory, shaped by empiricist and positivist philosophies.

Associated with the natural, applied, formal, and social sciences this research strategy promotes the objective empirical investigation of observable phenomena to test and understand relationships. This is done through a range of quantifying methods and techniques, reflecting on its broad utilization as a research strategy across different academic disciplines. The objective of quantitative research is to develop and employ mathematical models, theories, and hypotheses about phenomena. The process of measurement is central to quantitative research because it provides the fundamental connection between empirical observation and the mathematical expression of quantitative relationships.



System Requirement Specifications

Project Planning - On the first phase of the System Development Life Cycle. The proponents brainstormed possible places where they could conduct the study. After a deliberate discussion, the proponents decided to conduct the said study in Barangay Gulod Health Center. Possible problems to encounter as well as the future requirements needed to develop the Patient Information System for Barangay Gulod Health Center were also discussed.

Requirement Analysis. On the second phase, the proponents gathered information about the needs of the health center by conducting an interview with the Officer in Charge and health workers of Barangay Gulod Health Center .

Designing. On the third phase, the proponents started designing the expected output of the system by displaying the system flow, as well as its data, operations, and interactions using a flow charts and other diagrams.

Coding/Implementation. On the fourth phase, the programmers began coding and building the system, as well as connecting the database and executing all of the essential outputs that require data from the database, once the Graphical User Interface (GUI) had been built.

Testing. On the fifth phase, to ensure that the program was completely functional, the team has conducted quality assurance testing. If any potential issue or defect was found, the developers would address them immediately. When all of this was finished, the final iteration of the product would be placed into production.



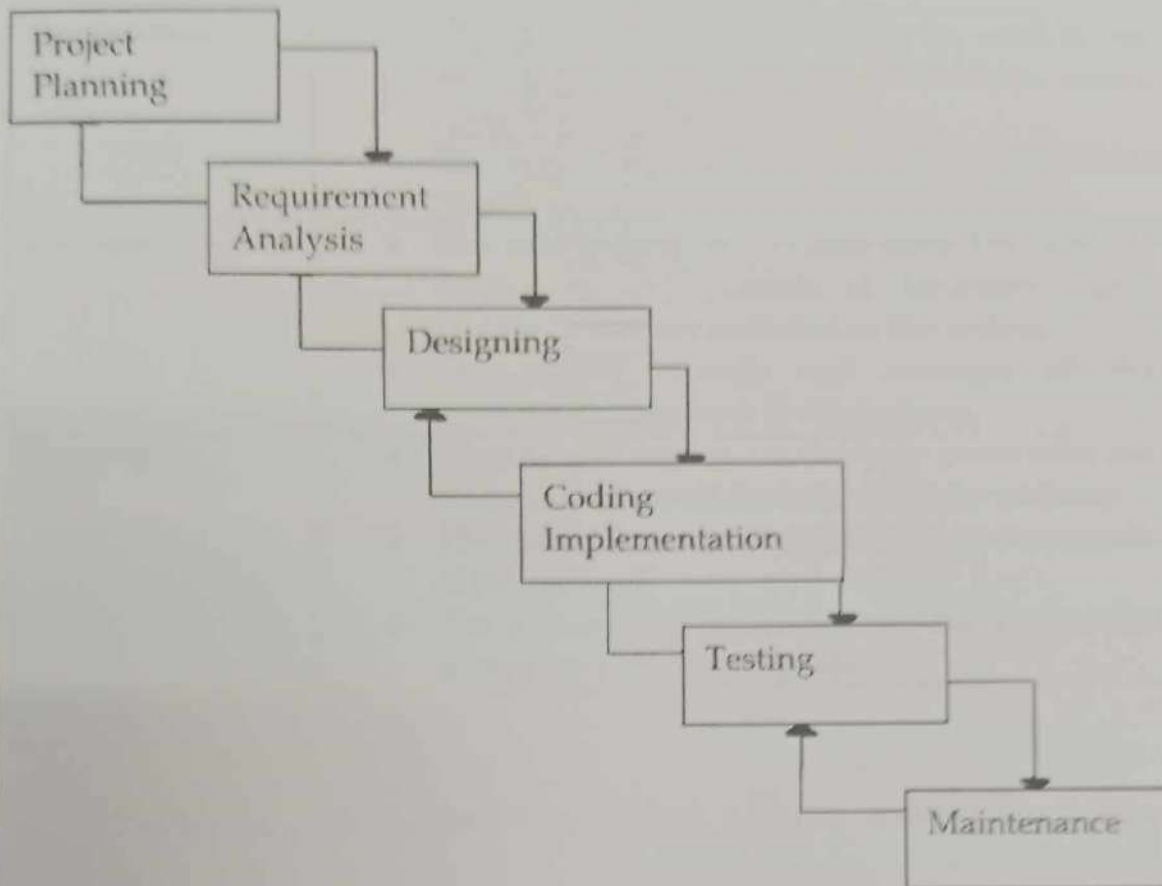
Maintenance. On the sixth phase, to ensure that our system was functioning well, maintenance would be performed once the unit was delivered and deployed to the client. If there were suggestions from the client, the system would be updated with a new feature or function.

System Development Methodologies

These are the phases involved in the development of the Patient Information System for Barangay Gulod Health Center as shown in Figure 3.1. The researchers selected the Modified Waterfall Model for the development of the Patient Information System for Barangay Gulod Health Center. Modified Waterfall Model is one of the models of System Development Life Cycle (SDLC) that includes a series of steps that involve the number of phases or procedures of how documents and systems are done.



Figure 3.1 Modified Waterfall Model of Patient Information System for
Barangay Gulod Health Center



Requirement Analysis

The researcher gathered information and identified the elements of the present system utilized by Barangay employees such as doctors, nurses, Midwives, and health workers through observation and interviews. Every day, patient information is recorded at Barangay Gulod Health Center, thus, the researchers presented a system that records and manages patient data and information, with features and requirements that include:

**Table 3.1 Requirement Analysis**

Requirement Type:	Example
Operational	• The system should be able to be used by the Barangay employees such as doctors, nurses, midwives, and health workers. • The System is only available at Barangay Gulod Health Center.
Performance	• The interactions and transactions between the employees and patients of Barangay Gulod Health Center are included in the system. • The system records and manages all the information and data of the patients.
Security	• Doctors and nurses are the only users who can manage and see all the records of the patients. • The system has a backup data for all the records of the patients. • The system has a log-in page for the doctors and nurses for security purposes.

Table 3.1. Shows the requirement analysis needed, including the requirement types for the implementation of the system.

Sampling technique

The proponent used convenience sampling to ensure that the sample adequately represented each group within the population. It also aids the proponents in obtaining a larger number of responses because participants came from a diverse range of backgrounds. As a primary source, the technique was based on the available population. Convenience sampling is a popular sample



technique because it is faster, simpler, and less expensive. Participants are also generally available to participate.

Because testing the entire population is practically impossible, convenience sampling is the most appropriate method for this project. When additional inputs are not available, researchers can use convenience sampling to monitor opinions in the simplest way possible, which is essential for primary study. In convenience sampling, no inclusion criteria are established prior to the selection of respondents, and all participants are welcome to participate.

The Slovin's Formula:

$$n = N / 1 + Ne^2$$

Where:

N = the size of the population

n = the size of sample

e = level of significance (0.08)

$$n = \frac{62,542}{1 + (0.08)^2}$$

$$n = 62,542 / 507.59$$

$$n = 156$$



Current Technical Situation

The Barangay Gulod Health Center's current process starts with the registration of patients in a first come, first served basis. The patients need to fill-out a registration book with their name, sex, age, address, and health concern. The health worker in charge of registration will find the records of old patients on a shelf allocated for storing the envelopes containing the records of all patients. Health workers will get basic information of new patients., then patients will be given a queue number. Once their number, they can see now the doctor in charge to examine them.

The health worker will assign a specific family number that they called family coder, that code is visible to their family envelope and each patient has their records in a bond paper containing their information, it is also known as ITR (individual treatment record). The ITR contains the patient's complete name, sex, age, birthday, address, height, weight, temperature, previous diagnosis, and their Phil health ID or PWD ID if they have one. Pregnancy and Family Planning patients have different shelves specifically for them, while TB patients' records are encoded in a specific computer. The information of patients can be seen by the health workers, nurses, and specific specialized doctors.

A patient record for Barangay Gulod health center is a repository of information about a single patient which is dedicated to collecting, storing, managing, and making available clinical information. The developer included a



local database for backup protection as well as an online database for easy control and sharing via the internet.

Requirement Documentation

The Patient Information System for Barangay Gulod Health Center is designed to collect, record, update, view, monitor, and manage patient information that can be accessible through a website.

In - Scope.

Login and Logout Module - This module allows users to enter their username and password to access the system.

Admin Module - The accounts of doctors, health workers, and patients in the system that may print or generate identity cards (IDs) are managed by this module.

This module may also provide the Barangay Gulod Health Center an overall report of patient records.

Health workers Module - This module keeps records of patient profiles, registers and updates patients for appointments, and provides notifications about their transactions at the Barangay Gulod Health Center.



Patient Module - This module manages the client profile and views medical history regarding their transaction in the Barangay Gulod Health Center.

Doctor Module - This module manages the schedule of patients, recording of medical findings and patients complaints and viewing of patient medical history.

Development and Testing

The researchers used Bootstrap as the front-end framework of the web application and PHP for the back-end side of the Patient Information System for Barangay Gulod Health Center. The researchers used Bunifu and Guna for the interface and C# for the back-end of the desktop application.

The System development and testing were used to evaluate quality of the system based on ISO 25010 quality model. The ISO 25010 standard was used to check the system software during development and testing stages. This elaborates the scope of development and testing under 5 classifications:

Functional Suitability

This characteristic represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions. This characteristic is composed of the following sub-characteristics:



- Functional Completeness – the degree to which the set of functions covers all the specified tasks and user objectives.
- Functional correctness – the degree to which a product or system provides the correct results with the needed degree of precision
- Functional appropriateness – the degree to which the functions facilitate the accomplishment of specified tasks and objectives.

Reliability

The degree to which a system, product, or component performs specific functions under specified conditions for a specified period. This characteristic is composed of the following sub-characteristics:

- **Maturity** – the degree to which a system, product, or component meets needs for reliability under normal operation.
- **Availability** – the degree to which a system, product or component is operational and accessible when required for use.
- **Fault tolerance** – the degree to which a system, product or component operates as intended despite the presence of hardware or software faults.
- **Recoverability** – the degree to which, in the event of an interruption or a failure, a product or system can recover the data directly affected and



re-establish the desired state of the system.

Usability

The degree to which a product or system can be used by specified users to achieve specified goals with effectiveness, efficiency and satisfaction in a specified context of use. This characteristic is composed of the following sub-characteristics.

- **Appropriateness recognizability** – the degree to which users can recognize whether a product or system is appropriate for their needs.
- **Learnability** – the degree to which a product or system can be used by specified users to achieve specified goals of learning to use the product or system with effectiveness, efficiency, freedom from risk and satisfaction in a specified context of use.
- **Operability** – the degree to which a product or system has attributes that make it easy to operate and control.
- **User error protection** – the degree to which a system protects users against making errors.
- **User interface aesthetics** – the degree to which a user interface enables pleasing and satisfying interaction for the user.
- **Accessibility** – the degree to which a product or system can be used by people with the widest range of characteristics and capabilities to achieve a specified goal in a specified context of use.



Performance Efficiency

This characteristic represents the performance relative to the number of resources used under stated conditions. This characteristic is composed of the following sub-characteristics:

- **Time behavior** – the degree to which the response and processing times and through put rates of a product or system, when performing its functions meet requirements.
- **Resource utilization** – the degree to which the amount and type of resources used by a product or system, when performing its functions, meet the requirements
- **Capacity** – the degree to which the maximum limits of a product or system parameter meet requirements.

Security

The degree to which a product or system protects information and data so that users, other products or systems have the degree of data access appropriate to their types and levels of authorization. This characteristic is composed of the following sub-characteristics:

- **Confidentiality** – the degree to which a product or system ensures that data are accessible only to those authorized to have access.



- **Integrity** - the degree to which a system, product or component prevents unauthorized access to, or modification of, computer programs or data.
- **Non-repudiation** - the degree to which actions or events can be proven to have taken place so that the events or actions cannot be repudiated later.
- **Accountability** - the degree to which the actions of an entity can be traced uniquely to the entity.
- **Authenticity** - the degree to which the identity of a subject or resource can be proved to be the one claimed.

Data Analysis Plan

The primary research instrument was survey questionnaire. The proponents used weighted mean as a statistical method to evaluate the Patient Information System for Barangay Gulod Health Center in accordance with the ISO 25010 criteria. Functionality, reliability, usability, efficiency, and security were used as criteria to evaluate the system. The mean scores of responses from survey questionnaires were calculated to determine the average of responses on the evaluation based on ISO 25010 criteria.



1. Frequency and Percentage Distribution

Frequency Distribution is the itemized breakdown of answers to each question depending on the study's categories or the set of responses defined in the survey.

Formula:

$$P = \frac{f}{N} \times 100\%$$

Where:

P = Percent

f = Frequency

N = Number of Respondents

2. Weighted Mean

To measure effectiveness, this rating scale was created utilizing a five-point Likert scale approach, with responses choosing one of the five potential reactions from the choices for each question. The weighted mean formula is used to get the average sum of the proposed system:

Formula:

$$WM = \frac{\sum fw}{n}$$

Where:

WM = weighted mean



Σ = summation

f = frequency

w = assigned weight

n = number of respondents

Likert scale was used and its interpretation as follows:

SCALE	INTERPRETATION	MEAN RANGE
5	Strongly Agree	4.21 - 5.00
4	Agree	3.41 - 4.20
3	Undecided	2.61 - 3.40
2	Disagree	1.81 - 2.60
1	Strongly Disagree	1.00 - 1.80

Table 3.2 Likert Scale

Table 3.2 shows the psychometric scale that the researchers used to understand the perspective of the respondents based on the evaluation of the system.

SOFTWARE CRITERIA	5	4	3	2	1
A. FUNCTIONALITY					
1. The system meets the requirements of the health care system.					
2. The system performs the required operations					
3. The system has the ability to handle specific business process.					
4. All modules are working in accordance to the company's requirements					
5. Functions in the system were well integrated					
B. RELIABILITY					
1. Conformance to the desired result					
2. Absence of failure					



requires only minimal amount of time. Finally, the system design was translated into machine code.

Table 3.4 Implementation Plan

STRATEGY	ACTIVITIES	PERSON INVOLVED	DURATION
APPROVAL FROM THE CITY HEALTH DEPARTMENT OFFICER - IN - CHARGE	LETTER FOR THE CITY HEALTH DEPARTMENT OFFICER - IN - CHARGE, GET THE APPROVAL MEMO	DR. ESPERANZA ARIAS, MR. JAYSON DELOS SANTOS	1 WEEK AND 3 DAYS
SYSTEM INSTALLATION	INSTALLATION OF THE SYSTEM AND REQUIRED HARDWARE AND SOFTWARE	PROGRAMMERS AND TECHNICALS	
INFORMATION DISTRIBUTION	DATA GATHERING	MEDICAL STAFF AND RESIDENT / PATIENTS OF THE HEALTH CENTER IN BRGY. GULOD	1 WEEK AND 5 DAYS
	RECORD INTERVIEW		
	TRANSCRIPT		
TESTING AND FINALIZATION OF THE RESEARCH	TESTING OF THE SYSTEM AND CHECKING OF THE DOCUMENTATION	PROGRAMMERS, TECHNICALS AND RESEARCHERS	

Table 3.4. shows the Implementation Plans that were used by the proponents. This table is the strategic plan used to accomplish the objectives of the research.



CHAPTER IV

RESULTS AND DISCUSSION

This chapter provides the results of statistical data relative to the objectives posted in chapter 1. The corresponding analysis and interpretation of data are incorporated in this portion of the study.

DEMOGRAPHIC PROFILE OF THE RESPONDENTS

Below are the tables that show the Frequency and Percentage distribution of the respondents Frequency and Percentage distribution by Age.

The proponents gathered the age of each respondent in order to identify age brackets who underwent Patient Information system for Barangay Gulod.

Table 4.1 Age of the respondents

Age	Frequency	Percentage	Rank
10 - 19	32	8.06%	5
20 - 29	78	19.64%	3
30 - 39	75	18.89%	4
40 - 49	115	28.96%	1
50 - 59	97	24.43%	2
Total	397	100%	

Th shows the evaluation of the respondents in the demographic profile of the respondents according to age. 8% of the respondents answer in the evaluation of 10-19 years old. 19% answered in the evaluation 20 -29 years old. 18% answered in the evaluation 30 -39 years old. The majority of the respondents belong to the



age bracket 40-49 years old. That is 28% since these respondents are the major users of the Patient Information system in barangay Gulod. And 24% answered in 50 - 59 ages of respondents.

Position in health center	Frequency	Percentage	Rank
Doctor	1	0.2%	2
Nurse	1	0.2%	2
Dentist	1	0.2%	2
Mid Wife	1	0.2%	2
Health Workers	0	0	3
Patient	393	98.99%	1
Total	397	100%	

Table 4.2 Respondents' position in health center

This table shows the evaluation of the respondents in the demographic profile according to respondent's position in health center. The majority of respondents are patients with a percentage of 99%, while doctors, nurses, dentists, and midwives have the same frequency of 1 a percent, and health workers have a frequency of 0 or 0 percent.

Assessment of the Patient Information System for Barangay Gulod

According to Criteria

This section determines whether the system was created in conformity with the prescribed criteria and standards. An assessment was conducted giving the



respondents of the study a freedom to rate the system being, truthful and objective as possible.

Table 4.3

Respondents' assessment of the proposed system in terms of Functionality in question number 1 "The system meets the requirements of the health care center."

Likert Scale	Frequency	Percentage	Rank
Strongly Agree	165	41.56%	2
Agree	177	44.58%	1
Undecided	45	11.34%	3
Disagree	10	2.52	4
Strongly Disagree	0	0.00%	5
Total	397	100%	

The table 4.3 shows that respondents answered "Strongly Agree" with the frequency of 165 (41.56%), while "Agree" has frequency of 177 (44.58%), "Undecided" with frequency of 45 (11.34%) , "Disagree" has frequency of 10 (2.52%) and "Strongly Disagree" has frequency of 0 (0.00%). In total, 397 people responded.

**Table 4.4**

Respondents' assessment of the proposed system in terms of Functionality in question number 2 "The system performs the required operations"

Likert Scale	Frequency	Percentage	Rank
Strongly Agree	131	33.0%	2
Agree	212	53.40%	1
Undecided	52	13.10%	3
Disagree	2	0.50%	4
Strongly Disagree	0	0.00%	5
Total	397	100%	

The table 4.4 shows that respondents answered "Strongly Agree" with the frequency of 131 (33.0%), while "Agree" has frequency of 212 (53.40%), "Undecided" with frequency of 52 (13.10%), "Disagree" has a frequency of 2 (0.50%). "Strongly Disagree" has a frequency of 0 (0.00%) In total of 397 people have responded.

**Table 4.5**

Respondents' assessment of the proposed system in terms of Functionality in question number 3 "The system has the ability to handle specific business process."

Likert Scale	Frequency	Percentage	Rank
Strongly Agree	161	40.55%	2
Agree	191	48.11%	1
Undecided	42	10.58%	3
Disagree	3	0.76%	4
Strongly Disagree	0	0.00%	5
Total	397	100%	

The table 4.5 shows that respondents answered "Strongly Agree" with the frequency of 161 (40.55%) ,while "Agree" has frequency of 191 (48.11%), "Undecided" with frequency of 42 (10.58%), "Disagree" has a frequency of 3 (0.76%), and "Strongly Disagree" has a frequency of 0(0.00%) In total, 397 people have responded.

**Table 4.6**

Respondents' assessment of the proposed system in terms of Functionality in question number 4 "All modules are working in accordance to the company's requirements"

Likert Scale	Frequency	Percentage	Rank
Strongly Agree	137	34.51%	2
Agree	175	44.08%	1
Undecided	84	21.16%	3
Disagree	1	0.25%	4
Strongly Disagree	0	0.00%	5
Total	397	100%	

The table 4.6 shows that respondents answered "Strongly Agree" with the frequency of 137 (34.51%), while "Agree" has frequency of 175 (44.08%), "Undecided" with frequency of 84 (21.16%) "Disagree" has the a frequency of 1 (0.25%). And "Strongly Disagree" has a total of 0 (0.00%) In total of 397 people have responded.

**Table 4.7**

Respondents' assessment of the proposed system in terms of Functionality in question number 5 "Functions in the system were well integrated."

Likert Scale	Frequency	Percentage	Rank
Strongly Agree	177	44.58%	1
Agree	167	42.07%	2
Undecided	48	12.09%	3
Disagree	5	1.26%	4
Strongly Disagree	0	0.00%	5
Total	397	100%	

The table 4.7 shows that respondents answered "Strongly Agree" with the frequency of 177 (44.58%), while "Agree" has frequency of 167 (42.07%), "Undecided" with frequency of 48 (12.09%), "Disagree" has a frequency of 5 (1.26%) and "Strongly Disagree" has frequency of 0 (0.00%). In total, 397 people have responded.

**Table 4.8**

Respondents' assessment of the proposed system in terms of Reliability in question number 1 "Conformance to the desired result."

Likert Scale	Frequency	Percentage	Rank
Strongly Agree	161	40.55%	2
Agree	180	45.34%	1
Undecided	54	13.60%	3
Disagree	2	0.50%	4
Strongly Disagree	0	0.00%	5
Total	397	100%	

The table 4.8 shows that respondents answered "Strongly Agree" with the frequency of 161 (40.55%), while "Agree" has frequency of 180 (45.34%), "Undecided" with frequency of 54 (13.6%), "Disagree" has a frequency of 2(0.50) "Strongly Disagree" has a frequency of 0 (0.00%). In total, 397 people have responded.

**Table 4.9**

Respondents' assessment of the proposed system in terms of Reliability in question number 2 "Absence of failure"

Likert Scale	Frequency	Percentage	Rank
Strongly Agree	155	39.04%	2
Agree	163	41.06%	1
Undecided	78	19.65%	3
Disagree	1	0.25%	4
Strongly Disagree	0	0.00%	5
Total	397	100%	

The table 4.9 shows that most respondents answered "Strongly Agree" with the frequency of 155 (39.04%), while "Agree" has frequency of 163 (41.06%), "Undecided" with frequency of 78 (19.65%), "Disagree" has a frequency of 1(0.25%) and "Strongly Disagree" has a frequency of 0 (0.00%). In total, 397 people have responded.



Table 4.10

Respondents' assessment of the proposed system in terms of Reliability in question number 3 "Accuracy in performance"

Likert Scale	Frequency	Percentage	Rank
Strongly Agree	163	41.06%	2
Agree	205	51.64%	1
Undecided	24	6.05%	3
Disagree	5	1.26%	4
Strongly Disagree	0	0.00%	5
Total	397	100	

The table 4.10 shows that most respondents answered "Strongly Agree" with the frequency of 163 (41.06%), while "Agree" has frequency of 205 (51.64%), "Undecided" with frequency of 24 (6.05%), "Disagree" has frequency of 5 (1.26%) and "Strongly Disagree" has frequency of 0 (0.00%). In total of 397 have people responded.

**Table 4.11**

Respondents' assessment of the proposed system in terms of Reliability in question number 4 "I can count on information I can get from this system"

Likert Scale	Frequency	Percentage	Rank
Strongly Agree	151	38.04%	1
Agree	140	35.26%	2
Undecided	104	26.20%	3
Disagree	2	0.50%	4
Strongly Disagree	0	0.00%	5
Total	397	100%	

The table 4.11 shows that respondents answered "Strongly Agree" with the frequency of 151 (38%), while "Agree" has frequency of 140 (35.26%), "Undecided" with frequency of 104 (26.2%), "Disagree" has a frequency of 2(0.50%) and "Strongly Disagree" has a frequency of 0 (0.00%). In total, 397 have people responded.

**Table 4.12**

Respondents' assessment of the proposed system in terms of Reliability in question number 5 "I feel confident doing transactions using this system"

Likert Scale	Frequency	Percentage	Rank
Strongly Agree	179	45.09%	1
Agree	163	41.06%	2
Undecided	53	13.35%	3
Disagree	2	0.50%	4
Strongly Disagree	0	0.00%	5
Total	397	100%	

The table 4.12 shows that respondents answered "Strongly Agree" with the frequency of 179 (45.0%), while "Agree" has frequency of 163 (41%), "Undecided" with frequency of 53 (13.35%), "Disagree" has a frequency of 2 (0.50) and lasty "Strongly Disagree" has a frequency of 0 (0.00%). In total, 397 have people responded.

**Table 4.13**

Respondents' assessment of the proposed system in terms of Usability in question number 1 "It was simple to use the system"

Likert Scale	Frequency	Percentage	Rank
Strongly Agree	249	62.72%	1
Agree	104	26.20%	2
Undecided	36	9.07%	3
Disagree	8	2.02%	4
Strongly Disagree	0	0.00%	5
Total	397	100%	

The table 4.13 shows that respondents answered "Strongly Agree" with the frequency of 249 (62.72%), while "Agree" has frequency of 104 (26.20%), "Undecided" with frequency of 36 (9.07%), "Disagree" has a frequency of 8 (2.02%) and lastly "Strongly Disagree" has a frequency of 0 (0.00%). In total, 397 people have responded.

**Table 4.14**

Respondents' assessment of the proposed system in terms of Usability in question number 2 "It was easy to learn to use the system"

Likert Scale	Frequency	Percentage	Rank
Strongly Agree	216	54.41%	1
Agree	144	36.27%	2
Undecided	36	9.07%	3
Disagree	1	0.25%	4
Strongly Disagree	0	0.00%	5
Total	397	100%	

The table 4.14 shows that respondents answered "Strongly Agree" with the frequency of 216 (54.41%), while "Agree" has frequency of 144 (36.27%), "Undecided" with frequency of 36 (9.07%), "Disagree" has a frequency of 1 (0.25) and lastly "Strongly Disagree" has a frequency of 0 (0.00%). In total, 397 people have responded.

**Table 4.15**

Respondents' assessment of the proposed system in terms of Usability in question number 3 "Helps the user become more productive"

Likert Scale	Frequency	Percentage	Rank
Strongly Agree	170	42.82%	2
Agree	180	45.34%	1
Undecided	38	9.57%	3
Disagree	9	2.27%	4
Strongly Disagree	0	0.00%	5
Total	397	100%	

The table 4.15 shows that respondents answers "Strongly Agree" with the frequency of 170 (42.82%), while "Agree" has frequency of 180 (45.34%), "Undecided" with frequency of 38 (9.57%), "Disagree" has a frequency of 9 (2.27%) and lasty "Strongly Disagree" has a frequency of 0 (0.00%). In total, 397 people have responded.

**Table 4.16**

Respondents' assessment of the proposed system in terms of Usability in question number 4 "The organization of information on screen is clear"

Likert Scale	Frequency	Percentage	Rank
Strongly Agree	191	48.11%	1
Agree	155	39.04%	2
Undecided	48	12.09%	3
Disagree	3	0.76%	4
Strongly Disagree	0	0.00%	5
Total	397	100	

The table 4.16 shows that respondents answered "Strongly Agree" with the frequency of 191 (48.11%), while "Agree" has frequency of 155 (39.04%), "Undecided" with frequency of 48 (12.09%), "Disagree" has a frequency of 3 (0.76) and lasty "Strongly Disagree" has a frequency of 0 (0.00%). In total, 397 people have responded.



Table 4.17

Respondents' assessment of the proposed system in terms of Usability in question number 5 "The interface of the system is pleasant"

Likert Scale	Frequency	Percentage	Rank
Strongly Agree	182	45.84%	1
Agree	154	38.79%	2
Undecided	59	14.86%	3
Disagree	2	0.50%	4
Strongly Disagree	0	0.00%	5
Total	397	100%	

The table 4.17 shows that respondents answered "Strongly Agree" with the frequency of 182 (45.84%), while "Agree" has frequency of 154 (38.79%), "Undecided" with frequency of 59 (14.86%), "Disagree" has a frequency of 2 (0.50) and lasty "Strongly Disagree" has a frequency of 0 (0.00%). In total, 397 people have responded.

**Table 4.18**

Respondents' assessment of the proposed system in terms of Efficiency in question number 1 "Process are completed with less time and effort"

Likert Scale	Frequency	Percentage	Rank
Strongly Agree	235	59.19%	1
Agree	144	36.27%	2
Undecided	17	4.28%	3
Disagree	1	0.25%	4
Strongly Disagree	0	0.00%	5
Total	397	100	

The table 4.18 shows that respondents answered "Strongly Agree" with the frequency of 235 (59.19%), while "Agree" has frequency of 144 (36.27%), "Undecided" with frequency of 17 (4.28%), "Disagree" has a frequency of 1 (0.25) and lasty "Strongly Disagree" has a frequency of 0 (0.00%). In total, 397 people have responded.

**Table 4.19**

Respondents' assessment of the proposed system in terms of Efficiency in question number 2 "Helps the user effectively complete their work"

Likert Scale	Frequency	Percentage	Rank
Strongly Agree	165	41.56%	2
Agree	210	52.90%	1
Undecided	22	5.54%	3
Disagree	0	0.00%	4
Strongly Disagree	0	0.00%	4
Total	55	100%	

The table 4.19 shows that respondents answered "Strongly Agree" with the frequency of 165 (41.56%), while "Agree" has frequency of 210 (52.90%), "Undecided" with frequency of 22 (5.54%), and "Disagree, and Strongly Disagree has the same frequency of 0 (0.00%) In total, 397 people have responded.

**Table 4.20**

Respondents' assessment of the proposed system in terms of Efficiency in question number 3 "Helps the users become productive"

Likert Scale	Frequency	Percentage	Rank
Strongly Agree	204	51.39%	1
Agree	130	32.75%	2
Undecided	61	15.37%	3
Disagree	2	0.50%	4
Strongly Disagree	0	0.00%	5
Total	397	100%	

The table 4.20 shows that respondents answered "Strongly Agree" with the frequency of 204 (51.39%), while "Agree" has frequency of 130 (32.75%), "Undecided" with frequency of 61 (15.37%), "Disagree" has a frequency of 2 (0.50) and lasty "Strongly Disagree" has a frequency of 0 (0.00%). In total, 397 people have responded.

**Table 4.21**

Respondents' assessment of the proposed system in terms of Security in question number 1 "The system protects your information against any malicious attack "

Likert Scale	Frequency	Percentage	Rank
Strongly Agree	225	56.68%	1
Agree	134	33.75%	2
Undecided	38	9.57%	3
Disagree	0	0.00%	4
Strongly Disagree	0	0.00%	4
Total	397	100%	

The table 4.21 shows that respondents answered "Strongly Agree" with the frequency of 225 (56.68%), while "Agree" has frequency of 134 (33.75%), "Undecided" with frequency of 38 (9.57%), and "Disagree, and Strongly Disagree has the same frequency of 0 (0.00%) In total, 397 people have responded.



Table 4.22

Respondents' assessment of the proposed system in terms of Security in question number 2 "The system is well-built with back up information."

Likert Scale	Frequency	Percentage	Rank
Strongly Agree	218	54.91%	1
Agree	143	36.02%	2
Undecided	32	8.06%	3
Disagree	4	1.01%	4
Strongly Disagree	0	0.00%	5
Total	397	100%	

The table 4.22 shows that respondents answered "Strongly Agree" with the frequency of 218 (54.91%), while "Agree" has frequency of 143 (36.06%), "Undecided" with frequency of 32 (8.06%), "Disagree" has a frequency of 4 (1.01%) and lasty "Strongly Disagree" has a frequency of 0 (0.00%). In total, 397 people have responded.

**Table 4.23**

Respondents' assessment of the proposed system in terms of Security in question number 3 "The system ensures that data are accessible only to those authorized to have access. "

Likert Scale	Frequency	Percentage	Rank
Strongly Agree	185	46.60%	1
Agree	138	34.76%	2
Undecided	73	18.39%	3
Disagree	1	0.25%	4
Strongly Disagree	0	0.00%	5
Total	397	100%	

The table 4.23 shows that respondents answered "Strongly Agree" with the frequency of 185 (46.60%), while "Agree" has frequency of 138 (34.76%), "Undecided" with frequency of 73 (18.39%), "Disagree" has a frequency of 1 (0.25) and lasty "Strongly Disagree" has a frequency of 0 (0.00%). In total, 397 people have responded.



Summary of Assessment of the respondents on the Criteria

This portion summarizes the respondents of the system according to criteria used with the corresponding weighted mean and interpretation for functionality, reliability, efficiency, usability and security.

Table 4.24

Summary Assessment for Functionality

Statements	5(Strongly Agree)	4(Agree)	3(Undecided)	2(Disagree)	1(Strongly Disagree)	Mean	Verbal Interpretation
Does the system meet the requirements of the company?	165	177	45	10	0	4.25	Strongly Agree
Does required operations are performed by the developed system?	131	212	52	2	0	4.21	Strongly Agree
Does the system have the ability to handle specific business process?	161	191	42	3	0	4.28	Strongly Agree



All modules are working in accordance to the company's requirements.	137	175	84	1	0	4.12	Agree
Functions in the system were well integrated.	177	167	48	5	0	4.29	Strongly Agree
						Weighted Mean = 4.23	Strongly Agree

This portion summarizes the respondents of the system according to criteria used with the corresponding weighted mean and interpretation for reliability.



Table 4.25

Summary Assessment for Reliability

Statements	5(Strongly Agree)	4(Agree)	3(Undecided)	2(Disagree)	1(Strongly Disagree)	Mean	Verbal Interpretation
Conformance to the desired result.	161	180	54	2	0	4.25	Strongly Agree
Absence of failure	155	163	78	1	0	4.18	Agree
Accuracy in performance	163	205	24	5	0	4.32	Strongly Agree
I can count on information I can get from this system	151	140	104	2	0	4.1	Agree
I feel confident doing transactions using this system	179	163	53	2	0	4.3	Strongly Agree
						Weighted Mean = 4.23	Strongly Agree



This portion summarizes the respondents of the proposed system according to criteria used with the corresponding weighted mean and interpretation for efficiency.

Table 4.26

Summary Assessment for Efficiency

Statements	5(Strongly Agree)	4(Agree)	3(Undecided)	2(Disagree)	1(Strongly Disagree)	Mean	Verbal Interpretation
Process is completed with less time and effort	235	144	17	1	0	4.54	Strongly Agree
Helps the user effectively complete their work	165	210	22	0	0	4.54	Strongly Agree
Helps the users become productive	204	130	61	2	0	4.35	Strongly Agree
						Weighted Mean = 4.41	Strongly Agree



This portion summarizes the respondents of the proposed system according to criteria used with the corresponding weighted mean and interpretation for usability.

Table 4.27**Summary Assessment for Usability**

Statements	5(Strongly Agree)	4(Agree)	3(Undecided)	2(Disagree)	1(Strongly Disagree)	Mean	Verbal Interpretation
It was simple to use the system	249	104	36	8	0	4.49	Strongly Agree
It was easy to learn to use the system	216	144	36	1	0	4.4	Strongly Agree
Helps the user become more productive	170	180	38	9	0	4.28	Strongly Agree
The organization of information on screen is clear	191	155	48	3	0	4.34	Strongly Agree



The interface of the system is pleasant	182	154	59	2	0	4.29	Strongly Agree
						Weighted Mean = 4.36	Strongly Agree

This portion summarizes the respondents of the proposed system according to criteria used with the corresponding weighted mean and interpretation for security.

Table 2.28

Summary Assessment for Security

Statements	5(Strongly Agree)	4(Agree)	3(Undecided)	2(Disagree)	1(Strongly Disagree)	Mean	Verbal Interpretation
The system protects your information against any malicious attack.	225	134	38	0	0	4.47	Strongly Agree
The system is well-	218	143	32	4	0		



built with back-up information.						4.44	Strongly Agree
The system ensures that data are accessible only to those authorized to have access.						4.27	Strongly Agree
	185	138	73	1	0		
						Weighted Mean = 4.39	Strongly Agree

Summary of Assessment of the respondents on the proposed system

This portion summarizes the respondents of the proposed system according to criteria used with the corresponding weighted mean and over-all interpretation

Assessment Summary of the respondents on the Proposed System.

The data shows that the researchers had set five (5) criteria for the respondents to assess the system. Based on the computation made, table shows that the weighted mean formula the criteria is equivalent to Functionality,



Reliability, Usability, Efficiency, and Security and the average weighted mean is 4.32.

Criteria	Weighted Mean	Verbal Interpretation	Rank
Functionality	4.23	Strongly Agree	1
Reliability	4.23	Strongly Agree	1
Usability	4.36	Strongly Agree	1
Efficiency	4.41	Strongly Agree	1
Security	4.39	Strongly Agree	1
Average Weighted Mean	4.32	Strongly Agree	

Table 4.29

The table shows "Strongly Agree" as an overall interpretation. It clearly suggests that the proposed system has met all the criteria set forth. Indeed, the proposed system is found to be Functional, Reliability, Usability Efficiency, and Security.



CHAPTER V

SUMMARY OF FINDINGS, CONCLUSION, AND RECOMMENDATIONS

This chapter offers a summary of the findings and recommendations achieved by the researchers during this project. The findings are reported to indicate how effective the system is, and suggestions were made to improve the project's future development, for future researchers and the health care centers.

Summary of findings

Based on the gathered data, carefully interpreted and analyzed, below are the summarized findings:

1. Profile of the Respondents

Age. The age bracket of the respondents are as follows: 10 – 14, 20 – 29, 30 – 39, 40 – 49, 50 – 59. An age bracket of 10-14 years old has 8.06% of the respondents. An age bracket of 20 -29 years old has 19.64% answered in the evaluation. For the age bracket of 30 -39 years old has 18.89% of the respondent. The majority of the respondents belong to the age bracket of 40-49 years old has 28.96% And the remaining respondents of an age bracket of 50 -59 years old has 24.43% only.



The proponents also include the age of respondents in the demographic profile so that we can determine what age they prefer to use the system, and the age at which they easily adapt to the process

Respondent's position in health center. out of total 397 respondents, The Doctor, Nurse, Dentist, Midwife has the same percentage and make up of 0.2% of the respondents. The majority of respondents is the Patients and has 98.99% of it.

2. Assessment of respondents on the proposed system according to the following criteria:

In terms of functionality, the patient information system for Brgy Gulod health center received strongly agree results with an overall weighted mean of **4.23**. In terms of reliability, the patient information system for Brgy Gulod health center received strongly agree results with an overall weighted mean of **4.23**. In terms of efficiency, the patient information system for Brgy Gulod health center received strongly agree results with an overall weighted mean of **4.41**. In terms of usability, the patient information system for Brgy Gulod health center received strongly agree results with an overall weighted mean of **4.36**. In terms of portability, the patient information system for Brgy Gulod health center received strongly agree results with an overall weighted mean of **4.39**.

The primary goal of this research is to improve the methods for the barangay gulod health center and make them more efficient for both health



workers and patients. The data gathered will be used to build a system that will reflect all of the research, observation, and interviews conducted by the proponents. The proponents evaluate the patient information for the barangay gulod health center based on the ISO25010 criteria of functionality, reliability, usability, efficiency, and security. Based on the evaluation, the proponents receive an average weighted mean of 4.32, with the verbal interpretation of strongly agree based on the likert scale interpretation used by the proponents.

Conclusion

The following conclusions are based on the findings this study:

1. Respondents felt unsafe in crowded areas, particularly health centers, during this pandemic. The health center obtained information from patients using a manual or paper-based approach which is time-consuming, unproductive, and unrecoverable. Regardless, a patient must consult a doctor due to disease and a lack of funds at a public health center such as Barangay Gulod is still valuable to the residents.
2. Using the Patient Information System, it was possible to have a more convenient way of getting patient information or records of the residents of the Barangay Gulod compared to the previous manual system. Following



the system evaluation, implementing the system will be essential for improving the patient records process in terms of:

- Functional Suitability - functions that meet stated and implied needs when used under specified conditions such as completeness, accuracy, and appropriateness.
- Reliability is defined as the ability of a system, p or component to execute given functions under specified conditions over a set amount of time.
- Usability - entails providing a user-friendly and simple-to-use application.
- Performance Efficiency - the ability to complete a task with minimal human intervention.
- Security - entails ensuring and protecting the user's sensitive information and integrity through the use of authentication.

3. Health is an essential component of an individual's life. The participation of each individual in undertaking a self-health evaluation and assuming responsibility for it, will boost sickness prevention. As a result, the researchers conclude that a



4. Patient Information System can improve cost control, increase the efficiency, effectiveness, and accuracy of patient safety, improve service capacity, reduce personnel costs, and improve healthcare quality, all of which will benefit health workers and residents of Barangay Gulod.
5. In terms of functionality, dependability, usability, efficiency, and security, the proposed project received a satisfactory score in all of these areas. As we can see, the goals and objectives were met. As a result, the researchers are confident that if the system is implemented by the Barangay Gulod Health Center, it will be highly beneficial to both the staff and the residents of Barangay Gulod. In carrying out their jobs with ease, as well as providing their patients with access to their health information.

Recommendation

Based on the findings and conclusions presented, the following recommendations are suggested:

1. Since the findings revealed that the system is very functional, reliability, usability, efficient and secure, it is therefore recommended that Patient Information System will be used and implement in Barangay Gulod Health Center.
2. The proponents recommend that future researchers possibly integrate data analytics with graphs to monitor common diseases.



3. The proponents recommend that future researchers possibly add a "Recommender System" that will automatically suggest possible solutions to the most common diseases.
4. The proponent recommends adding an "Inventory System" for the medical supply of Barangay Gulod Health center for additional features.
5. The proponent recommends adding an "Queueing System" for additional features.
6. The proponents recommended that users of the system have knowledge in managing patient information and computer operation.
7. The proponents recommended upgrading the equipment to avoid any problem with the system performance.
8. The proponents suggest that the Barangay Gulod Health Center improve the queuing process in the proposed system by installing hardware like a kiosk printer for printing the queue number.
9. The proponents recommend Barangay Gulod Health Center consider having a cloud database to store data.



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APPENDICES

APPENDIX A

Request letter to the company

 **QUEZON CITY UNIVERSITY**
COLLEGE OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY
673 Quirino Highway, San Bartolome, Novaliches, Quezon City 

03/25/2022

Dr. Esperanza Anita N. Escañó-Arias, MD, MPH
OIC, City Health Department Head
Batasan Social Hygiene Clinic
ISP Road, Brgy. Batasan Quezon City

RECEIVED
MAR 25 2022

Dear Dr. Escañó-Arias,

The 4th year students of the program Bachelor of Science in Information Technology are required to survey and evaluate the system and capstone project. The Capstone Project that they will develop should be based on their observed problem/s and gathered information from their research locale.

A Capstone Project Group chose your company as one of the stakeholders to their plan **Web-based Patient Information System (WPIS) for Barangay Gulod: A Recommendation System and Health Information System**. Our students are excited to work with your company, and allowing them to survey or evaluate the system for the completion of the project is a big help.

Allow me to provide you the possible activities the Capstone group shall exercise.

- The evaluation of the system of the Capstone Project group will start upon your permission.
- Confidentiality and ethics are already oriented to the students but help us remind them about security measures on accessing data and documents from your company.

Your company's contributions to our students' explorations and experiences in conducting research and developing capstone projects will be appreciated.

If there are matters that you want me to discuss with you, please contact me thru this email address edwin.macaspag@qcu.edu.ph

Thank you for your kind consideration.

Very truly yours,


MR. EDWIN MACASPAG
Adviser, Capstone Project and Research I

Noted by:
ISAGANI M. TANO, DIT
Dean, College of CS and IT